

MONJED

Web-Based Robot Simulation

Concept, Look and Feel, Simulated Workstreams, and Budget Brief

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Audience: product, engineering, partners, and funders

Related live systems: API (Render), Frontend (Vercel)

This brief describes a browser-based robot / drone mission simulator that visualizes how MONJED turns disaster risk into coordinated field action - without requiring physical robots for demos, training, or planning.

1. Purpose

MONJED already connects early warning (flood / earthquake risk), decisions, community help requests, volunteers, and alerts. The robot simulation extends that story into autonomous or semi-autonomous field assets: robots and drones that can be tasked when risk rises or when people request help.

The simulation is not a full robotics operating system (not Gazebo / ROS replacement). It is a web mission simulator: map, agents, MONJED decisions, and outcomes - designed for demos, training, and scenario planning.

Goals

- Show the full loop: hazard -> MONJED risk/decision -> robot task -> status update
- Train operators and partners without hardware
- Support fundraising and partner demos on monjed.vercel.app
- Optionally write mission records into the same admin / ops views as real dispatches

2. How it will look (UX)

Primary surface: a full-page Simulation screen inside the MONJED web app (same visual language as Map / Admin / Volunteer dashboards).

Layout (desktop)

- Left / center (dominant): interactive map (Leaflet or MapLibre) with flood/quake zones, shelters, blocked roads, and robot icons
- Right panel: scenario controls (Play / Pause / Speed), injected hazard, selected robot details, linked help request, mission log
- Top bar: scenario name, risk level badge, connection mode (Live API vs Mock)
- Bottom ticker (optional): alert / SMS / dispatch events for storytelling

Visual language

- Robots: simple markers with status color (idle / en route / on site / returning / failed)
- Paths: animated routes from base to target (no heavy 3D required for v1)
- Hazards: translucent zone overlays consistent with existing risk map
- Missions: cards that mirror assistance / volunteer assignment UI

Mobile

Map-first with a bottom sheet for controls. Full multi-robot editing stays desktop-first.

v1 look summary

- 2D map simulation (not Unity/WebGL required)
- 1-3 robots + 1 flood scenario in a known MONJED zone
- Playable in the browser at monjed.vercel.app/simulation

3. What work it will simulate

The simulator models disaster-response field work that MONJED could coordinate - starting with software-defined missions that later can map to real robots.

Mission types (v1 focus)

- Reconnaissance: drone/robot surveys a flooded area after a HIGH risk decision
- Last-mile check-in: robot navigates toward a help-request location to confirm status
- Supply drop cue: simulate carrying water / first-aid markers to a shelter waypoint
- Hazard confirmation: validate community report evidence near a risk hotspot

Mission types (later)

- Multi-robot area coverage and battery constraints
- Blocked-route replanning when roads flood
- Failure modes: lost link, low battery, inaccessible target
- Human-in-the-loop: operator approve / reject before dispatch

What is intentionally NOT simulated in v1

- Full physics, LIDAR, SLAM, or motor control
- Real radio / cellular network modeling
- Hardware safety certification

Link to MONJED product work

- Uses same concepts as pipeline risk, assistance requests, volunteer matching, and alerts
- Can call live Render API or a mock decision engine for offline demos
- Can create mission records visible in admin ops views

4. System design (high level)

Components

- Frontend sim UI (React + map + agent tick loop)
- Simulation engine (client-side first): positions, ETA, status transitions
- MONJED API (optional live): risk/decision/assistance endpoints on Render
- Mission API (new, small): create/list/update simulated missions in MongoDB
- Scenario packs (JSON): map bounds, agents, waypoints, scripted events

Modes

- Mock mode: no backend required - ideal for demos and flaky networks
- Live mode: risk/decision from <https://monjed.onrender.com>
- Hybrid: live help requests + local robot motion

Suggested API additions (minimal)

- POST /simulation/missions - create a mission from a help request or scenario
- GET /simulation/missions - list missions for admin/sim UI
- PATCH /simulation/missions/{id} - update status (en_route, on_site, completed, failed)
- GET /simulation/scenarios - list packaged scenarios

Existing MONJED endpoints (pipeline, assistance, dashboard) remain the source of truth for risk and human help queues. Simulation should not fork that logic.

5. Delivery phases

Phase 0 - Concept lock (1 week)

- Agree scenarios, robot roles, and success demo script
- Wireframe Simulation page

Phase 1 - Web MVP (3-5 weeks)

- Map + 1 scenario + 1-3 agents + play/pause
- Mock decision path + mission log
- Deploy on Vercel next to existing frontend

Phase 2 - API integration (2-3 weeks)

- Mission endpoints + Mongo persistence
- Live mode against Render API
- Admin panel: view simulated missions

Phase 3 - Training / partner polish (2-4 weeks)

- 2-3 scenario packs, failure modes, replay
- Optional light 3D or richer animation if needed

Phase 4 - Hardware bridge (optional, separate budget)

- ROS / vendor SDK adapters outside the browser
- Only after web mission model is stable

6. Budget estimate

Figures are planning ranges in USD for a small team. They exclude physical robots unless noted. Local salaries vary by country; use the band that matches your hiring market.

A) Software-only web simulation (recommended start)

Total: about USD 8,000 - 25,000 (or 6-12 person-weeks)

- Product / UX design: USD 1,000 - 3,000
- Frontend simulation UI + map agents: USD 4,000 - 12,000
- Backend mission API + Mongo models: USD 1,500 - 5,000
- QA, scenarios, demo polish: USD 1,000 - 3,000
- Contingency (15%): included in upper bound

B) Hosting / API / tools (monthly, ongoing)

- Vercel frontend: USD 0 - 20 (hobby/pro)
- Render backend: USD 0 - 25 (free/starter; paid if always-on needed)
- MongoDB Atlas: USD 0 - 30 (free M0 often enough for sim MVP)
- Domains / monitoring / SMS sandbox extras: USD 0 - 20
- Monthly ops total (MVP): about USD 0 - 100

C) If you buy contractor time (example mix)

- 1 fullstack engineer at USD 25-60/hr x 160-240 hrs: USD 4,000 - 14,400
- Part-time designer at USD 20-40/hr x 20-40 hrs: USD 400 - 1,600
- PM / demo scripting: USD 500 - 2,000

7. Budget - optional extensions

D) Richer 3D web scene (optional)

- Three.js / light WebGL scene: add USD 3,000 - 10,000
- Only if partners require cinematic demos; not required for ops value

E) Physical robot pilot (separate from web sim)

- Education / hobby UGV or drone kit: USD 300 - 3,000 per unit
- Field-ready inspection drone: USD 2,000 - 15,000+
- Ground robot (research/light industrial): USD 5,000 - 50,000+
- Integration engineer (ROS / vendor SDK): USD 5,000 - 30,000

Recommendation: do not buy robots until the web mission model and MONJED dispatch UX are proven.

F) What you already have (cost avoided)

- MONJED frontend (React/Vite) deployed on Vercel
- MONJED API on Render
- Risk/decision/assistance domain model
- Map concepts already in product

Building simulation on top of this stack is cheaper than a greenfield robotics demo.

8. Risks and success criteria

Risks

- Overbuilding physics instead of mission storytelling
- Confusing partners that the browser agents are real robots
- API cold starts (Render free tier) hurting live demos - use Mock mode for shows
- Empty Mongo / missing env on deploy blocking login-related flows

Success criteria for MVP

- A partner can open /simulation and understand MONJED action loop in under 3 minutes
- One scripted flood scenario runs end-to-end without engineer help
- Mission states are visible and explainable
- Works on desktop Chrome/Edge; acceptable on tablet

9. Recommendation

Build a web-based 2D mission simulator first. Keep robots as map agents controlled by a simple state machine, driven by MONJED risk/decision/help-request concepts. Add a thin mission API only when you need persistence and admin visibility.

Budget to approve for a credible MVP: target USD 10,000-18,000 software (or equivalent internal sprint capacity), plus under USD 100/month hosting. Defer physical robots to a later pilot with a separate hardware budget.

Immediate next steps

- Approve Phase 1 scope (1 scenario, 1-3 agents, mock mode)
- Choose owner (frontend-led) and 2 demo scripts
- Add /simulation route on the existing Vercel frontend
- Keep Render API as optional live brain; Mongo mission collection when ready

End of brief. This document is a planning artifact for MONJED robot simulation and does not authorize procurement by itself.